Killing Rate (γ) vs. Degradation by Living R (φ_max): Is the Relation Linear?

Model: Ma et al. (2024) two-strain model with the paper's SI parameters · one dose · 22 × 21 × 2 = 924 simulations plus 11 iso-lines found by root-finding · same engine and endpoint definitions as the main parameter sweep report.

Open issues to solve next

- **γ is extended beyond the paper's range** (1.1–1.4 in the paper, 1.1–3.1 here; group choice, same as the killing vs. protection test).
- **Iso-lines are followed beyond the paper's φ_max range.** The paper uses 0–5; the lines are traced up to φ_max = 40 so their shape can be seen.
- **The linearity is empirical.** No formula was derived for it; it holds for this model over these ranges.
- **Plain model only.** No extinction floor; at high γ and dose both strains fall far below one cell before regrowing.
- **All other parameters at base.** With the inhibitor at i = 10 the effective degradation is 0.31 × φ_max.

Base parameters & variation range

Table 1. Variation levels.

Quantity	Meaning	Base	Range	Values	Source
γ (y-axis)	killing rate: maximum lysis coefficient	1.38	1.1–3.1	22	SI range 1.1–1.4, extended
φ_max (x-axis)	antibiotic degradation by living R	1	0–5	21	SI range
a0	antibiotic dose	2.15	2.15 and 10	2	base dose and a strongly killing dose

Decoupling of the parameters

- **Full factorial:** every γ is combined with every φ_max, for each dose. Each axis has 21 evenly spaced values (plus the base value for γ).
- **Only two parameters move.** α, β_min, c, κ_b, d_a and everything else stay at base.

Sweep setup:

- φ_max does not enter the equation for the R/S ratio. It acts only by shortening the time the antibiotic stays, so there is no closed-form neutral line as there was for protection.
- Iso-line: for each γ, the φ_max that holds one endpoint at a fixed level, found by root-finding on the simulation. It answers “how much more degradation offsets more killing?”.
- Endpoints held fixed: peak S loss at 50%, 90%, 99%, 99.99% and at its base-point value; final resistant fraction at its base-point value.
- Linearity is judged by the R^2 of a straight-line fit of φ_max against γ along each iso-line, and by the largest deviation from that line.

1. The γ × φ_max Plane

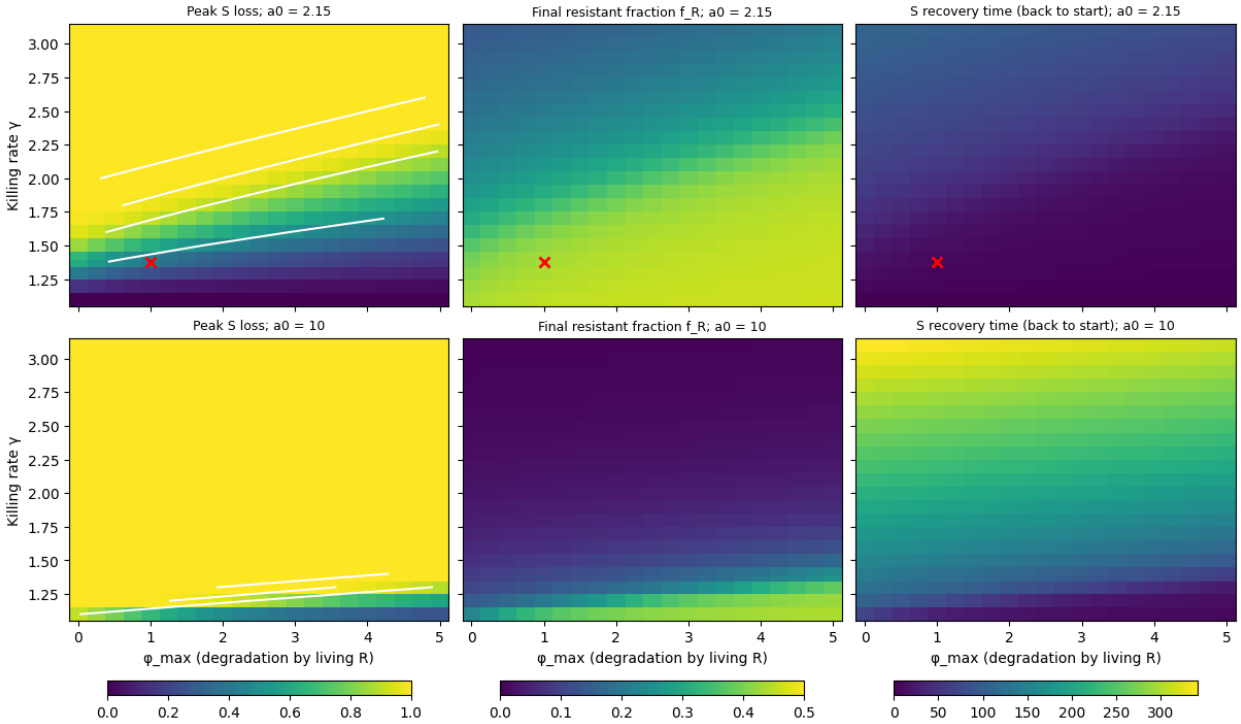


Figure 1. Peak S loss, final resistant fraction and S recovery time on the γ × φ_max plane. White lines: iso-lines of peak S loss (50%, 90%, 99%, 99.99%). Red cross: base point. Top: base dose. Bottom: a0 = 10.

- All 924 runs settle with both strains present (plain model).
- **γ has the longer reach at the base dose.** There γ takes peak S loss from 0% to 100% at every φ_max, while φ_max (0–5) moves it by at most 61 points (at γ = 1.6). At a0 = 10 the loss never drops below 26%, and γ (9–74 points) and φ_max (up to 65 points) have comparable reach.
- More degradation means less S loss, a higher final resistant fraction and a faster recovery, at every γ.
- The iso-lines run as straight, tilted bands across the plane.

2. Iso-Lines: φ_max Needed to Offset γ

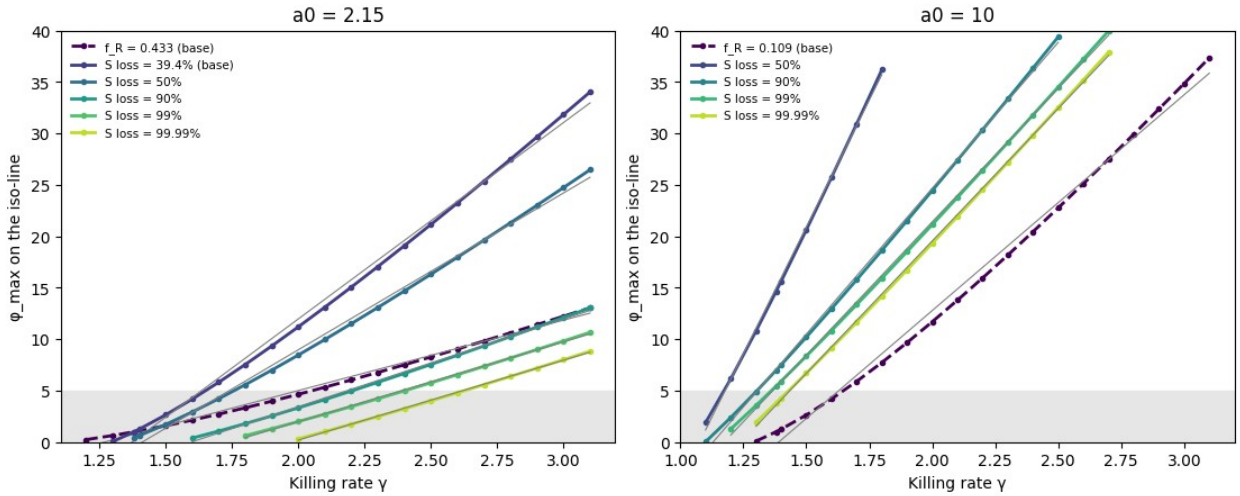


Figure 2. φ_max on each iso-line against γ. Solid: peak S loss levels. Dashed: final resistant fraction at its base value. Thin grey: straight-line fits. Shaded: the paper's φ_max range (0–5).

Table 2. Straight-line fits of φ_max against γ along each iso-line. Deviation = largest distance from the fitted line, as a share of the range. Slope ratio = slope at the high-γ end / slope at the low-γ end. Last column: width of the γ band the line covers inside φ_max 0–5.

Dose a0	Iso-line	Points	γ range	Slope	R^2	Deviation	Slope ratio	γ band in SI range
2.15	f_R = 0.433 (base)	21	1.2–3.1	6.8	0.9939	5.5%	2.05	0.91
2.15	S loss = 39.4% (base)	20	1.3–3.1	19.1	0.9961	4.2%	1.85	0.35
2.15	S loss = 50%	19	1.38–3.1	15.2	0.9968	3.4%	1.71	0.42
2.15	S loss = 90%	16	1.6–3.1	8.5	0.9983	2.8%	1.38	0.66
2.15	S loss = 99%	14	1.8–3.1	7.8	0.9991	1.9%	1.24	0.70
2.15	S loss = 99.99%	12	2–3.1	7.8	0.9995	1.4%	1.17	0.67
10	f_R = 0.109 (base)	20	1.3–3.1	21.0	0.9936	5.3%	2.20	0.35
10	S loss = 50%	9	1.1–1.8	49.3	0.9989	2.1%	1.24	0.07
10	S loss = 90%	16	1.1–2.5	28.4	0.9991	2.2%	1.28	0.20
10	S loss = 99%	17	1.2–2.7	25.9	0.9995	1.5%	1.19	0.22
10	S loss = 99.99%	16	1.3–2.7	25.7	0.9997	1.1%	1.15	0.21

- **Close to linear, with a slope that rises along each line.** A straight line stays within 1–6% of the range on all eleven lines (R^2 = 0.994–0.9997).
- Every line bends upward: the slope at the high-γ end is 1.2 to 2.2 times the slope at the low-γ end. Only the high-loss lines (99% and above) are close to straight.
- **The slope is steep.** At the base dose each +0.1 in γ needs on average +0.8 (99% loss line) to +1.9 (39% loss line) more φ_max. At a0 = 10 it needs +2.6 to +4.9.
- **Degradation offsets only a narrow band of killing rates.** Inside the paper's φ_max range an iso-line covers a γ band of 0.35–0.91 at the base dose and 0.07–0.35 at a0 = 10.
- The final resistant fraction behaves the same way: the φ_max that keeps it at its base value is a near-straight line in γ (R^2 = 0.994).

3. Effect of φ_max at a Fixed γ

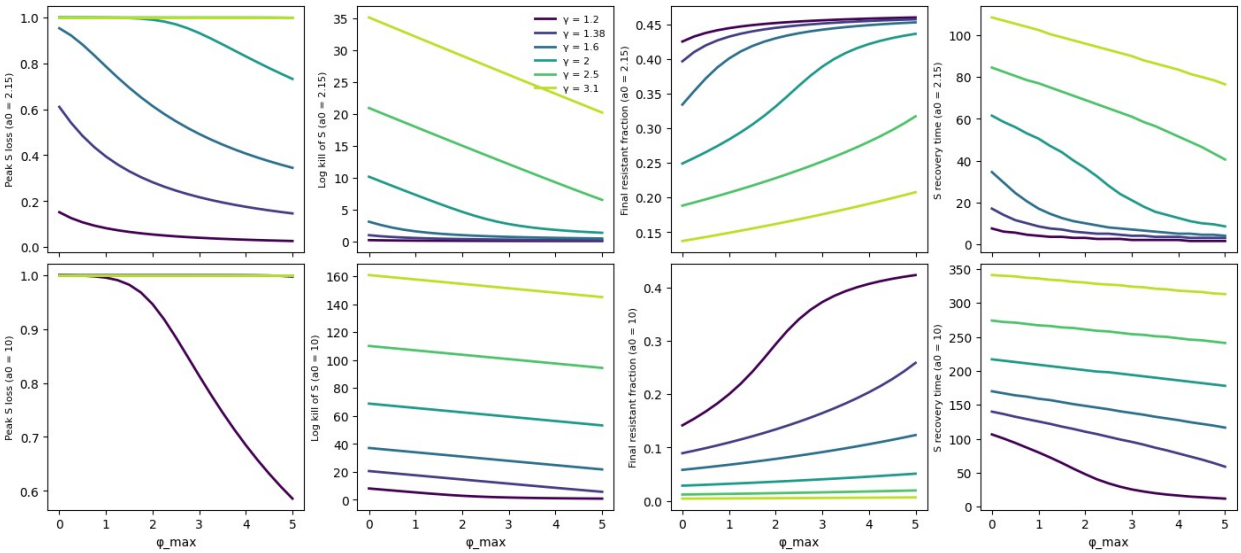


Figure 3. Peak S loss, log kill of S, final resistant fraction and S recovery time against φ_max, one line per γ. Top: base dose. Bottom: a0 = 10.

Table 3. R^2 of straight-line fits at fixed γ. Log kill = ln(starting density / lowest density).

Dose a0	γ	Log kill of S at φ_max = 0	Log kill vs φ_max	Log kill vs 1/(1 + φ_max)	Final f_R vs φ_max	Recovery time vs φ_max
2.15	1.2	0.2	0.809	0.999	0.826	0.815
2.15	1.38	0.9	0.816	0.998	0.795	0.782
2.15	1.6	3.1	0.797	0.998	0.794	0.787
2.15	2	10.1	0.932	0.917	0.978	0.974
2.15	2.5	20.9	1.000	0.796	0.987	0.995
2.15	3.1	35.1	1.000	0.791	0.997	1.000
10	1.2	8.1	0.882	0.951	0.951	0.929
10	1.38	20.6	1.000	0.791	0.972	0.997
10	1.6	37.1	1.000	0.788	0.990	1.000
10	2	68.9	1.000	0.788	0.994	1.000
10	2.5	110.2	1.000	0.788	0.996	0.999
10	3.1	160.9	1.000	0.788	0.996	0.999

- **Strong killing: linear.** When the crash is deep (log kill above about 20), the log kill of S falls in a straight line with φ_max (R^2 = 1.000), and so does the recovery time.
- **Weak killing: not linear.** The log kill follows 1/(1 + φ_max) more closely (R^2 = 0.998 at the base dose for γ ≤ 1.6), so the first unit of φ_max matters most.
- Peak S loss as a fraction saturates at 100% and is never a good linear measure; log kill is.

4. Conclusions

- **γ and φ_max are close to linearly related:** the degradation needed to offset more killing rises as a near-straight line in γ (R^2 ≥ 0.994), bending upward.

- **The exchange rate is unfavourable:** +0.1 in γ costs +0.8 to +4.9 in φ_max, against a φ_max range of only 0–5.
- **So φ_max shifts outcomes only inside a narrow band of γ;** at the base dose γ has the longer reach.
- **At fixed γ,** the crash depth is linear in φ_max under strong killing and follows 1/(1 + φ_max) under weak killing.
- Compared with private protection (previous test): protection trades off inversely with γ during treatment; degradation trades off almost linearly.
- Next: derive the iso-line slope analytically, repeat with κ_b and d_a, and add the extinction floor.